

UAV Simulation Description

The project simulates **fixed-wing UAVs** with realistic constraints like a **minimum turning radius**, managing multiple UAVs as they navigate to targets based on commands.

Key Components

1. UAV Class:

Models UAV behavior:

- **Attributes:** Position (x, y, z), speed, azimuth, turning radius, and target.
- **Methods:**
 - **setTarget:** Assigns a new target.
 - **update:** Updates position and heading:
 - Adjusts azimuth by moving along a **circular path (clockwise)** until aligned.
 - Moves straight toward the target after alignment.
 - Hover mode: Circles the target within a set radius.
 - **printStatus:** Logs UAV status to a file.

2. Simulation Class:

Manages multiple UAVs and executes commands.

- **Attributes:** Time step dt , time limit, UAV list, commands.
 - **Methods:**
 - **loadParams:** Reads simulation settings and initializes UAVs.
 - **loadCommands:** Parses and stores UAV movement commands.
 - **runSimulation:** Simulates UAV behavior step-by-step, executes commands, updates UAVs, and logs results.
-

Workflow

1. Setup:

- Load parameters and create UAVs.
- Load commands for target updates.

2. Simulation:

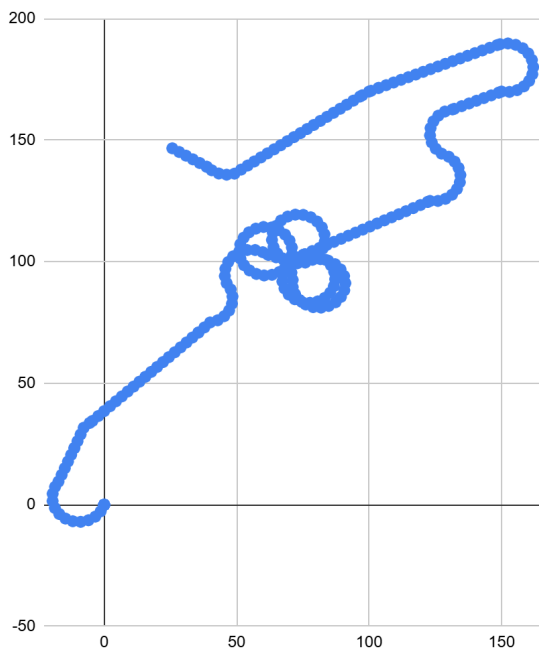
- At each step:
 - Align UAV azimuth using circular turning (clockwise).
 - Move straight toward targets or circle around them if reached.
 - Log UAV positions.

3. Output:

- Save UAV movement history to separate files (e.g., **UAV0.txt**).
-

Graphs

UAV0:



UAV1:

